

Member ID: _____

Time: _____

Rank: _____



MANAGERIAL ACCOUNTING (135) NATIONAL 2023

CONCEPT KNOWLEDGE:

Multiple Choice (20 @ 2 points each) _____ (40 points)

APPLICATION KNOWLEDGE:

Short Answer _____ (54 points)

Problem 1: Budgeted Income Statement _____ (51 points)

Problem 2: Flexible Budgets _____ (65 points)

Problem 3: Capital Budgets _____ (30 points)

TOTAL POINTS _____ (***240 points***)

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
2. Electronic devices will be monitored according to ACT standards.

The test is divided into two parts, and you **MUST** complete both parts. You will have 90 minutes to complete your work.

PART ONE Instructions: Concept Knowledge

1. Open Google Chrome and navigate to compete.bpa.org.
2. Click on the appropriate Post-Secondary Division Competitions button.
3. Enter your BPA Member ID (8-digits) and Password to access your test questions.
4. Answer all questions and submit your test.
5. Move on to Part Two.

PART TWO: Application Knowledge

Complete the questions in the rest of this test booklet by filling in the PDF.

EXAM GUIDELINES:

You have been hired as a Financial Assistant and will be keeping the accounting records for Digital Solutions, located at 700 Morse Road, Suite 201, Columbus, Ohio 43214. Digital Solutions provides accounting and other financial services for clients. You will complete jobs for Digital Solutions' own accounting records, as well as for clients.

The test is divided into two parts: concept knowledge and application knowledge.

Assumptions to make when taking this assessment:

- Round all calculations to two decimal places at the final step.
- Round all percentages to one decimal place.
- Use 360 days for interest calculations.

Multiple Choice

Identify the letter of the choice that *best* completes the statement or answers the question.

1. If budgets are to be effective, all of the following must be present except:
 - A. Acceptance at all levels of management
 - B. Research and analysis in setting realistic goals
 - C. Stockholders' approval of the budget
 - D. Sound organizational structure
2. If a company has adopted continuous budgeting, the budget will show plans for
 - A. Every day
 - B. The next 12 months
 - C. The current year and next year
 - D. The next quarter
3. An overly optimistic sales budget may result in
 - A. Increases in selling prices late in the year
 - B. Insufficient inventories
 - C. Increased sales during the year
 - D. Excessive inventories
4. The production budget shows expected unit sales of 32,000. Beginning finished goods inventory units are 5,600 and required production units are 33,600. What are the desired ending finished goods units?
 - A. 4,000
 - B. 5,600
 - C. 6,400
 - D. 7,200
5. Which of the following expenses would not appear on a selling and administrative expense budget?
 - A. Sales commissions
 - B. Depreciation
 - C. Property taxes
 - D. Indirect labor
6. Which of the following items would never appear on a cash budget?
 - A. Office salaries expense
 - B. Interest expense
 - C. Depreciation expense
 - D. Travel expense

7. Instead of a production budget, a merchandiser will prepare a
 - A. Merchandise purchases budget
 - B. Master time sheet
 - C. Sales forecast
 - D. Pseudo-production budget
8. A critical factor in budgeting for a service firm is to
 - A. Hire professional staff to perform the budgeting work
 - B. Coordinate professional staff needs with anticipated services
 - C. Classify all personnel as either variable or fixed
 - D. Budget expenditures before anticipated receipts
9. A static budget
 - A. Should not be prepared in a company
 - B. Is useful in evaluating a manager's performance by comparing actual variable costs and planned variable costs
 - C. Shows planned results at the original budgeted activity level
 - D. Is changed only if the actual level of activity is different than originally budgeted
10. Which one of the following will not increase return on investment?
 - A. Variable costs increase
 - B. Sales increase
 - C. Average operating assets decrease
 - D. Variable costs decrease
11. Which of the following would decrease residual income?
 - A. Decrease in required rate of return
 - B. Increase in the return on investment desired
 - C. Increase in sales
 - D. Increase in contribution margin
12. The difference between a budget and a standard is that
 - A. A budget expresses what costs were, while a standard expresses what costs should be.
 - B. A budget expresses management's plans, while a standard reflects what happened.
 - C. A budget expresses a total amount, while a standard expresses a unit amount.
 - D. Standards are excluded from the cost accounting system, whereas budgets are generally incorporated into the cost accounting system.

13. An unfavorable materials quantity variance would occur if
- A. More materials were purchased than were used
 - B. Actual pounds of materials used were less than the standard pounds allowed
 - C. The actual labor hours used were greater than the standard labor hours allowed
 - D. Actual pounds of materials used were greater than the standard pounds allowed
14. The costing of inventories at standard cost for external financial statement reporting purposes is
- A. Not permitted
 - B. Preferable to reporting on actual costs
In accordance with GAAP if significant differences exist between actual and standard costs
 - C. In accordance with GAAP if significant differences do not exist between actual and standard costs.
15. The direct materials price standard should include an amount for all of the following except
- A. Receiving costs
 - B. Storing costs
 - C. Handling costs
 - D. Normal spoilage costs
16. Which of the following is not relevant information in a decision whether old equipment presently being used should be replaced by new equipment?
- A. The cash price of the new equipment
 - B. The salvage value of the old equipment
 - C. The book value of the old equipment
 - D. The cost savings if the new equipment is purchased
17. A negative net present value means that the
- A. Project's rate of return exceeds the required rate of return
 - B. Project's rate of return is less than the required rate of return
 - C. Project's rate of return equals the required rate of return
 - D. Project is not profitable
18. If the net present value of a project is zero based on a discount rate of 16%, then the internal rate of return is:
- A. Equal to 16%
 - B. Less than 16%
 - C. Greater than 16%
 - D. Cannot be determined from this data

19. Lafleur Corporation is planning an investment with the following characteristics (Ignore income taxes):

Useful life	6 years
Yearly net cash inflow	\$45,000
Salvage value	\$0
Internal rate of return	18%
Required rate of return	14%

The initial cost of the equipment is closest to (See present value tables provided at the end)

- A. \$157,410
 - B. \$175,005
 - C. \$235,890
 - D. Cannot be determined from the given information
20. Jones Corporation is considering several investment proposals, as shown below:

	A	B	C	D
Investment required	\$80,000	\$100,000	\$60,000	\$75,000
Present value of future net cash flows	96,000	150,000	84,000	120,000

If the project profitability index is used, the ranking of the projects from most to least profitable would be:

- A. D, B, C, A
- B. B, D, C, A
- C. B, D, A, C
- D. A, C, B, D

Short Answer (54 points)

1. Renner company makes and sells dog harnesses. Each harness uses $\frac{3}{4}$ yard of nylon. Budgeted production of harnesses in units for the next three months is as follows:

	June	July	August
Budgeted production	21,000	20,000	24,000

The company wants to maintain monthly ending inventories of plastic equal to 25% of the following month's budgeted production needs; beginning inventory is the expected amount. The cost of nylon is \$2.12 per yard.

How many yards of nylon should Renner purchase in June?	
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What is the budgeted cost of materials for June?	
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2. Barton Inc. provided the following information:

	March	April	May
Projected merchandise purchases	\$76,000	\$65,000	\$70,000

Barton pays for 40% of merchandise purchases in the month purchased. General operating expenses are budgeted to be \$20,000 per month, of which \$2,000 is depreciation. Barton pays operating expenses in the month incurred.

What are Barton's cash disbursements for May?	
What will be the balance in Accounts Payable at the end of May?	

3. Cardellini Inc.'s manufacturing costs for May when production was 1,000 units appears below:

Direct materials	\$12,000
Direct labor	6,500
Variable overhead	5,000
Factory depreciation	9,000
Factory supervisor salaries	7,800
Other fixed factory costs	2,500

What is the flexible budget manufacturing cost for a month when 800 units are produced?	
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4. Steinfeld Company combines its operating expenses for budget purposes in a selling and administrative expense budget. For the first quarter, the following data are compiled:

Sales: 20,000 units; unit selling price:	\$35
Variable costs per dollar of sales:	
Sales commissions	6%
Delivery expense	2%
Advertising	4%
Fixed costs per quarter:	
Sales salaries	\$24,000
Office salaries	17,000
Depreciation	6,000
Insurance	2,000
Property taxes	1,000

What is the selling and administrative expense budget for the first quarter?	
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5. Whitfield Company produced and sold 50,000 units of its product and is operating at 70% of plant capacity. Each unit sells for \$70, and costs \$55 to produce. Fixed costs are 18% of the total manufacturing costs. Whitfield has received a proposal from another company to buy 15,000 units for \$50 per unit.

Should Whitfield accept the special order? Yes or no.	
What effect would acceptance of this order have on the company's income?	

6. Bartlet's Bikes has been manufacturing its own seats for its bicycles. The company is currently operating at 100% capacity, and variable manufacturing overhead is charged to production at the rate of 60% of direct labor cost. The direct materials and direct labor cost per unit to make the bicycle seats are \$8.00 and \$9.00, respectively. Normal production is 50,000 bicycles per year. A supplier offers to make the bicycle seats for \$22 each. If Bartlet accepts this offer, all variable manufacturing costs will be eliminated, but the \$30,000 of fixed manufacturing overhead currently being charged to the bicycle seats will have to be absorbed by other products.

Should Bartlet buy the seats from the outside supplier? Yes or no.	
What effect would buying the seats from the outside supplier have on the company's income?	

7. Olivia is the manager of Cat Castles, which manufactures climbing condos for cats. Cat Castles sells their condos to retailers unassembled for \$400. The cost of an unassembled condo is:

Direct materials	\$150
Direct labor	70
Variable overhead (70% of direct labor)	49
Fixed overhead (30% of direct labor)	<u>21</u>
Manufacturing cost per unit	<u>\$290</u>

Olivia has determined that their unused capacity is expected to continue indefinitely, and that some of that capacity can be used to assemble the condos and sell them for \$440. Assembling the condos will increase direct materials by \$5 per unit, and direct labor by \$10 per unit. Additional variable overhead will be incurred at the normal rates, but there is not expected to be an additional fixed overhead as a result of assembling the condos.

Should Cat Castles sell the condos assembled or unassembled?	
What effect would selling the condos assembled have on the company's income?	

8. Scantling Co. relies heavily on its 3D printer to print models of its products. The current printer is an older model and is inefficient. Management is considering updating the printer with a new model.

	Current Printer	New Model
Original purchase cost	\$10,000	\$20,000
Accumulated depreciation	8,000	
Estimated operating costs (annual)	9,000	4,200
Remaining useful life	5 years	5 years

If sold now, the current printer would have a salvage value of \$1,000; if operated for the remainder of its useful life, the salvage value is estimated to be zero. The new machine is expected to have a zero salvage value after five years.

Should Scantling buy the new machine or keep the old machine?	
What effect would buying the new machine have on income?	

9. Rodgers Inc. has three divisions: Green, Gold, White. The results of the first quarter of 2022 are presented below.

	Green	Gold	White	Total
Units sold	3,000	5,000	2,000	10,000
Revenue	\$70,000	\$50,000	\$40,000	\$160,000
Variable costs	32,000	26,000	16,000	74,000
Fixed costs	<u>20,000</u>	<u>32,000</u>	<u>16,000</u>	<u>68,000</u>
Net income	<u>\$18,000</u>	<u>(\$8,000)</u>	<u>\$8,000</u>	<u>\$18,000</u>

Allocated fixed costs are 43.75% of total fixed costs. Rodgers estimates that all the allocated costs will continue even if a division is discontinued. Since the Gold division is operating at a loss, Rodgers feels that it should be discontinued. If the Gold division is closed, it is estimated that sales of the Green division will increase by 10%, and sales at the White division will remain unchanged.

Should Rodgers discontinue the Gold division? Yes or no.	
What effect would discontinuing the Gold division have on the company's income?	

10. Crosby Company manufactures and sells two products. Relevant per unit data concerning each product is below.

	Standard	Deluxe
Selling price	\$28	\$32
Variable costs	\$10	\$12
Machine hours	4	5

If 1,000 additional machine hours are available, which product should be manufactured?	
If all the additional machine hours are allocated to the product you identified above, what would be the effect on total contribution margin?	

Problem 1: Budgeted Income Statement (51 points)

In September, the budget committee of Bishop Company assembles the following data:

- Expected Sales
 - October \$900,000
 - November \$850,000
 - December \$800,000
- Cost of goods sold is expected to be 60% of sales
- Desired ending inventory is 20% of next month’s cost of goods sold
- The beginning inventory on October 1 will be the desired amount

Prepare the budgeted income statement for October through gross profit, including a cost of goods sold schedule.

Problem 2: Flexible Budgets (65 points)

Lyman Company produces men's ties. Each tie is budgeted to use 1.5 yards of silk and 45 minutes of direct labor. In 2022, Lyman purchased 11,600 yards of silk and used 9,860, and Lyman's tailors worked a total of 4,220 hours. The following budgeted and actual amounts are for 2022:

	Budget at 5,000 units	Actual amounts at 5,800 units
Direct materials	\$60,000	\$70,992
Direct labor	75,000	86,510
Equipment depreciation (straight-line)	5,000	5,000
Indirect labor	7,500	8,600
Indirect materials	9,000	9,600
Rent & insurance	12,000	13,000

Prepare a budget report for Lyman Company for the year.

	Flexible Budget	Actual	Difference

What is the direct materials price variance?	
What is the direct materials usage variance?	
What is the direct labor rate variance?	
What is the direct labor efficiency variance?	

Problem 3: Capital Budgets (30 point)

Valdez Company is considering purchasing a new summer camp in the mountains of Wisconsin for \$3,000,000. The new facility will generate annual net cash inflows of \$500,000 for ten years. Specialists estimate that the facility will remain useful for ten years and have no residual value. The company uses straight-line depreciation, and its stockholders demand an annual return of 10% on investments of this nature.

Calculate the payback period of this investment	
Calculate the Accounting Rate of Return of this investment	
Calculate the Net Present Value of this investment	
Calculate the approximate Internal Rate of Return of this investment	
Calculate the Profitability Index of this investment	
Should Valdez invest in this project? Yes or no.	

Table A-1 Present Value of \$1

Present Value																
Periods	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	12%	14%	15%	16%	18%	20%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	0.893	0.877	0.870	0.862	0.847	0.833
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	0.797	0.769	0.756	0.743	0.718	0.694
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	0.712	0.675	0.658	0.641	0.609	0.579
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	0.636	0.592	0.572	0.552	0.516	0.482
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	0.567	0.519	0.497	0.476	0.437	0.402
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	0.507	0.456	0.432	0.410	0.370	0.335
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	0.452	0.400	0.376	0.354	0.314	0.279
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	0.404	0.351	0.327	0.305	0.266	0.233
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	0.361	0.308	0.284	0.263	0.225	0.194
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	0.322	0.270	0.247	0.227	0.191	0.162
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	0.287	0.237	0.215	0.195	0.162	0.135
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	0.257	0.208	0.187	0.168	0.137	0.112
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	0.229	0.182	0.163	0.145	0.116	0.093
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	0.205	0.160	0.141	0.125	0.099	0.078
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	0.183	0.140	0.123	0.108	0.084	0.065
16	0.853	0.728	0.623	0.534	0.458	0.394	0.339	0.292	0.252	0.218	0.163	0.123	0.107	0.093	0.071	0.054
17	0.844	0.714	0.605	0.513	0.436	0.371	0.317	0.270	0.231	0.198	0.146	0.108	0.093	0.080	0.060	0.045
18	0.836	0.700	0.587	0.494	0.416	0.350	0.296	0.250	0.212	0.180	0.130	0.095	0.081	0.069	0.051	0.038
19	0.828	0.686	0.570	0.475	0.396	0.331	0.277	0.232	0.194	0.164	0.116	0.083	0.070	0.060	0.043	0.031
20	0.820	0.673	0.554	0.456	0.377	0.312	0.258	0.215	0.178	0.149	0.104	0.073	0.061	0.051	0.037	0.026
21	0.811	0.660	0.538	0.439	0.359	0.294	0.242	0.199	0.164	0.135	0.093	0.064	0.053	0.044	0.031	0.022
22	0.803	0.647	0.522	0.422	0.342	0.278	0.226	0.184	0.150	0.123	0.083	0.056	0.046	0.038	0.026	0.018
23	0.795	0.634	0.507	0.406	0.326	0.262	0.211	0.170	0.138	0.112	0.074	0.049	0.040	0.033	0.022	0.015
24	0.788	0.622	0.492	0.390	0.310	0.247	0.197	0.158	0.126	0.102	0.066	0.043	0.035	0.028	0.019	0.013
25	0.780	0.610	0.478	0.375	0.295	0.233	0.184	0.146	0.116	0.092	0.059	0.038	0.030	0.024	0.016	0.010
26	0.772	0.598	0.464	0.361	0.281	0.220	0.172	0.135	0.106	0.084	0.053	0.033	0.026	0.021	0.014	0.009
27	0.764	0.586	0.450	0.347	0.268	0.207	0.161	0.125	0.098	0.076	0.047	0.029	0.023	0.018	0.011	0.007
28	0.757	0.574	0.437	0.333	0.255	0.196	0.150	0.116	0.090	0.069	0.042	0.026	0.020	0.016	0.010	0.006
29	0.749	0.563	0.424	0.321	0.243	0.185	0.141	0.107	0.082	0.063	0.037	0.022	0.017	0.014	0.008	0.005
30	0.742	0.552	0.412	0.308	0.231	0.174	0.131	0.099	0.075	0.057	0.033	0.020	0.015	0.012	0.007	0.004
40	0.672	0.453	0.307	0.208	0.142	0.097	0.067	0.046	0.032	0.022	0.011	0.005	0.004	0.003	0.001	0.001
50	0.608	0.372	0.228	0.141	0.087	0.054	0.034	0.021	0.013	0.009	0.003	0.001	0.001	0.001		

Table A-2 Present Value of Ordinary Annuity of \$1

Periods	Present Value															
	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	12%	14%	15%	16%	18%	20%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	0.893	0.877	0.870	0.862	0.847	0.833
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	1.690	1.647	1.626	1.605	1.566	1.528
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	2.402	2.322	2.283	2.246	2.174	2.106
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	3.037	2.914	2.855	2.798	2.690	2.589
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	3.605	3.433	3.352	3.274	3.127	2.991
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	4.111	3.889	3.784	3.685	3.498	3.326
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	4.564	4.288	4.160	4.039	3.812	3.605
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	4.968	4.639	4.487	4.344	4.078	3.837
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	5.328	4.946	4.772	4.607	4.303	4.031
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	5.650	5.216	5.019	4.833	4.494	4.192
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	5.938	5.453	5.234	5.029	4.656	4.327
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	6.194	5.660	5.421	5.197	4.793	4.439
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	6.424	5.842	5.583	5.342	4.910	4.533
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	6.628	6.002	5.724	5.468	5.008	4.611
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	6.811	6.142	5.847	5.575	5.092	4.675
16	14.718	13.578	12.561	11.652	10.838	10.106	9.447	8.851	8.313	7.824	6.974	6.265	5.954	5.669	5.162	4.730
17	15.562	14.292	13.166	12.166	11.274	10.477	9.763	9.122	8.544	8.022	7.120	6.373	6.047	5.749	5.222	4.775
18	16.398	14.992	13.754	12.659	11.690	10.828	10.059	9.372	8.756	8.201	7.250	6.467	6.128	5.818	5.273	4.812
19	17.226	15.678	14.324	13.134	12.085	11.158	10.336	9.604	8.950	8.365	7.366	6.550	6.198	5.877	5.316	4.844
20	18.046	16.351	14.877	13.590	12.462	11.470	10.594	9.818	9.129	8.514	7.469	6.623	6.259	5.929	5.353	4.870
21	18.857	17.011	15.415	14.029	12.821	11.764	10.836	10.017	9.292	8.649	7.562	6.687	6.312	5.973	5.384	4.891
22	19.660	17.658	15.937	14.451	13.163	12.042	11.061	10.201	9.442	8.772	7.645	6.743	6.359	6.011	5.410	4.909
23	20.456	18.292	16.444	14.857	13.489	12.303	11.272	10.371	9.580	8.883	7.718	6.792	6.399	6.044	5.432	4.925
24	21.243	18.914	16.936	15.247	13.799	12.550	11.469	10.529	9.707	8.985	7.784	6.835	6.434	6.073	5.451	4.937
25	22.023	19.523	17.413	15.622	14.094	12.783	11.654	10.675	9.823	9.077	7.843	6.873	6.464	6.097	5.467	4.948
26	22.795	20.121	17.877	15.983	14.375	13.003	11.826	10.810	9.929	9.161	7.896	6.906	6.491	6.118	5.480	4.956
27	23.560	20.707	18.327	16.330	14.643	13.211	11.987	10.935	10.027	9.237	7.943	6.935	6.514	6.136	5.492	4.964
28	24.316	21.281	18.764	16.663	14.898	13.406	12.137	11.051	10.116	9.307	7.984	6.961	6.534	6.152	5.502	4.970
29	25.066	21.844	19.188	16.984	15.141	13.591	12.278	11.158	10.198	9.370	8.022	6.983	6.551	6.166	5.510	4.975
30	25.808	22.396	19.600	17.292	15.372	13.765	12.409	11.258	10.274	9.427	8.055	7.003	6.566	6.177	5.517	4.979
40	32.835	27.355	23.115	19.793	17.159	15.046	13.332	11.925	10.757	9.779	8.244	7.105	6.642	6.234	5.548	4.997
50	39.196	31.424	25.730	21.482	18.256	15.762	13.801	12.233	10.962	9.915	8.304	7.133	6.661	6.246	5.554	4.999